

DUAL-LASER SECURITY SYSTEM

Using An Old Feature Phone

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A security system having a single laser beam may not be very effective as any moving object or animal can interrupt it, causing a false alarm. Presented here is a security system having two laser beams with calling alert feature, using a basic mobile phone (old model), also known as feature phone.

The security system can be used in a restricted area like a farm or as

a home security system. When an intruder cuts or interrupts both laser beams at the same time, an alarm sound is generated and an alert call is made to your current mobile phone.

Circuit and working

Circuit diagram of the laser security system using a basic feature phone is shown in Fig. 1. It is built around two light-dependent resistors

(LDR1 and LDR2), five transistors (T1 through T5), one diode (D1), one NE555 timer (IC1) and a few other components.

T3 and T4 are configured as NAND gate. NE555 is configured in monostable mode. T5 is used to drive relay RL1 and piezo buzzer PZ1. Relay contacts are connected to the old basic feature phone through connector CON2.

When two laser beams falling on LDR1 and LDR2 are interrupted at the same time, T1 and T2 conduct, making T3 and T4 also to conduct. Output signal at collector of T3 is given to trigger pin 2 of IC1 through toggle switch S1. As a result, LED1 goes off and T5 conducts.

PZ1 sounds for a pre-fixed time based on the values of resistor R8 and capacitor C1. At the same time, RL1 gets energised and activates the speed dial function of the feature phone through CON2. The speed dial function is as-

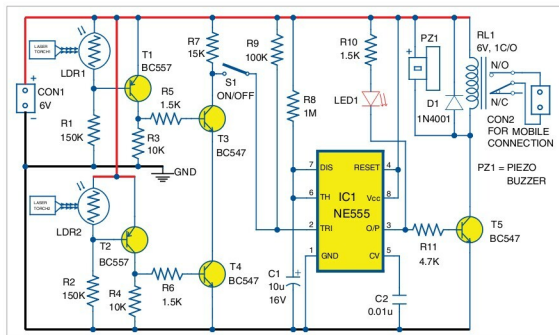


Fig. 1: Circuit diagram of the laser security system

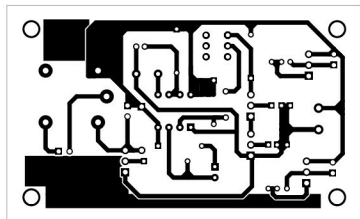


Fig. 2: Actual-size PCB layout of the laser security system

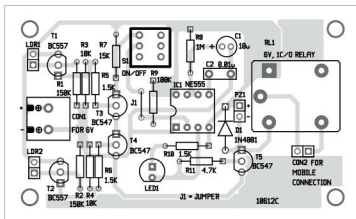


Fig. 3: Component layout of the PCB